Correlated Country Risk and International Capital Allocation

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Abstract

This paper studies how geography shapes international capital allocation and cross-country interest-rate differences. We first document a persistent geographic gradient in real interest rates: geographically distant countries exhibit larger bilateral interest-rate differentials, and this relationship remains in the persistent component of interest-rate gaps. We develop a multi-country portfolio-allocation model in which geography operates through two distinct channels: bilateral investor–destination portfolio frictions and spatially correlated destination attractiveness. Holding capital stocks fixed, the aggregate geographic contribution to interest-rate dispersion is driven primarily by spatially correlated destination attractiveness, although this masks substantial heterogeneity across country groups: geographic forces reduce interest-rate dispersion relative to the United States among advanced economies but increase it among emerging market economies. In steady state, both geographic channels have substantial effects on the cross-country allocation of capital and output, and removing both considerably compresses their distributions. These results show that geography matters for international finance not only through bilateral barriers to investment but also through spatial dependence across investment destinations.

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1 Introduction

Geographic distance continues to shape international investment despite substantial reductions in the technological cost of moving capital across borders. Investors disproportionately hold domestic and nearby foreign assets, and bilateral portfolio positions decline with distance. These patterns are commonly interpreted as evidence of information frictions, familiarity, or other barriers to international investment. Yet geography may matter for international capital allocation in another way. Countries located near one another may also share persistent characteristics that make them similar investment destinations. These two forms of geographic dependence are observationally related but economically distinct.

This paper studies how these two geographic forces shape international capital allocation and cross-country interest-rate differences. We begin by documenting that bilateral interest-rate differentials increase with geographic distance. The relationship remains when we isolate the persistent country-pair component of interest-rate differences after removing country-specific time-varying shocks. Thus, the geographic pattern in interest rates is not solely a consequence of transitory regional or macroeconomic fluctuations.

We distinguish two channels through which geography can generate this pattern. The first operates between an investor and a destination. Investors face bilateral portfolio frictions that increase with geographic distance, so that otherwise similar investment opportunities receive less capital when they are farther from the investor. The second operates across destinations themselves. Destination-specific characteristics that affect investment demand may be spatially correlated, so that geographically proximate countries exhibit similar attractiveness to investors from all origins. The distinction is economically important. For example, a U.S. investor may allocate less capital to Korea than to Canada because Korea is farther from the United States. This is a bilateral investor–destination friction. At the same time, Korea may share persistent destination characteristics with nearby countries such as Japan. This second form of geographic dependence does not depend on the location of any particular investor: investors from different origins may view nearby destinations as similar because their underlying investment characteristics are spatially correlated.

To quantify these channels, we develop a multi-country production economy with endogenous international portfolio allocation. Households save through competitive financial intermediaries

that allocate wealth across domestic and foreign destinations. Portfolio demand depends on the destination return, a bilateral portfolio wedge, and a destination-specific attractiveness component common to investors from all origins. On the production side, capital has a decreasing marginal product, so international portfolio allocation and equilibrium interest rates are jointly determined.

We estimate the bilateral portfolio wedge using international portfolio holdings. The portfolio-allocation equation generates a gravity relationship in which holdings decline with bilateral distance after controlling for investor-by-time and destination-by-time factors. The estimated coefficient on log distance is approximately -0.80, implying that doubling bilateral distance is associated with about a 42 percent decline in predicted bilateral portfolio holdings. We also find substantial home bias and lower portfolio holdings across geopolitical blocs.

Because equilibrium returns are jointly determined with portfolio allocation, we estimate the sensitivity of portfolio demand to returns using an instrumental-variable strategy that builds on Koijen and Yogo (2026). The instrument compares predicted demand pressure with predicted asset supply. Predicted demand is constructed from the estimated bilateral gravity structure using demand originating from investors other than the focal investor, while predicted supply is based on observable destination characteristics. The resulting variation in relative demand pressure shifts equilibrium returns through market clearing and identifies the return sensitivity of portfolio demand.

We next study the destination-side channel. We recover destination-specific attractiveness and first show that geographically proximate countries tend to have more similar destination components. This relationship remains after removing observable differences in market size and is not eliminated by controlling for geopolitical distance. We then decompose the residual destination component into a persistent spatial component, a persistent non-spatial country component, and transitory variation. The spatial covariance is modeled as a decreasing function of bilateral distance and estimated jointly with the remaining variance components by restricted maximum likelihood. The estimates indicate substantial persistent spatial dependence in destination attractiveness, with the spatial component exceeding the non-spatial component of persistent cross-country heterogeneity.

We use the estimated model to quantify the implications of these two geographic channels. We first hold country-level capital stocks fixed and decompose population-weighted interest-rate dispersion relative to the United States. For the full sample, baseline dispersion is 1.44 percentage points, compared with 0.76 percentage points when both geographic channels are removed. Most of the

aggregate increase is associated with the spatial component of destination attractiveness, while the aggregate main effect of bilateral geographic frictions is close to zero. This aggregate result, however, masks substantial heterogeneity. Geographic forces reduce interest-rate dispersion relative to the United States among advanced economies but increase it among emerging market economies, with the latter effect driven primarily by the spatial destination component.

We then allow capital stocks and equilibrium returns to adjust endogenously. In the general-equilibrium counterfactuals, removing either bilateral distance frictions or the spatial destination component reduces cross-country dispersion in capital and output. Removing both channels produces substantially larger reductions: the variance of log capital falls by about 64 percent and the variance of log GDP per capita by about 25 percent relative to the baseline. The quantitative importance of geography is more clearly reflected in the allocation of capital and output than in the equilibrium dispersion of interest rates alone.

These results show that geography in international finance is not well described by a single bilateral distance friction. Geographic distance affects the allocation of capital between investors and destinations, but investment destinations themselves also exhibit persistent spatial dependence. The relative importance of these channels differs across outcomes and country groups. Spatial destination characteristics account for most of the aggregate geographic contribution to interest-rate dispersion, while both channels play quantitatively important roles in determining the international distribution of capital and output.

Our paper contributes to the literature in three ways. First, we connect the geographic structure of international portfolio holdings to persistent cross-country interest-rate differences in a general-equilibrium model. Second, we distinguish bilateral investor–destination frictions from spatial dependence across destinations and estimate the latter directly from the covariance structure of destination attractiveness. Third, we quantify how the two channels affect interest-rate dispersion and the allocation of capital and output, highlighting substantial heterogeneity between advanced and emerging market economies.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 documents the geographic gradient in bilateral interest-rate differentials and its persistent component. Section 4 develops the model. Section 5 estimates bilateral portfolio frictions and the sensitivity of portfolio demand to returns. Section 6 estimates and recovers the spatial component of destination attrac-

tiveness. Section 7 presents the decomposition and general-equilibrium counterfactuals. Section 8 concludes.

Related Literatures Our paper relates to several strands of the literature. First, it contributes to the open-economy macroeconomics literature studying international capital allocation and global portfolio positions, including Lucas (1990), Obstfeld and Rogoff (1995), Gourinchas and Jeanne (2006), Pavlova and Rigobon (2007), Gourinchas and Rey (2007), Caselli and Feyrer (2007), Alfaro et al. (2008), Pellegrino et al. (2025), and Kleinman et al. (2026). Our analysis is also related to models of segmented international financial markets, including Gabaix and Maggiori (2015), Maggiori (2017), and Itskhoki and Mukhin (2021, 2025), which emphasize frictions that prevent capital from being allocated frictionlessly across countries.

Second, our analysis relates to the literature documenting gravity-type patterns in international capital allocation. Portes et al. (2001) and Portes and Rey (2005) show that international financial transactions decline with geographic distance, with related evidence provided by Aviat and Coeurdacier (2007) and Lane and Milesi-Ferretti (2008). Coeurdacier and Martin (2009) and Okawa and van Wincoop (2012) incorporate gravity-type portfolio relationships into models of international asset allocation. More recently, Koijen and Yogo (2026) exploit gravity-type variation in asset demand to construct instruments for estimating the sensitivity of international portfolio demand to returns. Our paper builds on this literature by treating geographic distance not only as a bilateral portfolio friction but also as a source of spatial structure in destination-side investment attractiveness.

Third, the paper relates to work studying spatial dependence in macroeconomics and finance, including LeSage and Llano (2013), Dell'Erba et al. (2013), Lustig and Richmond (2020), Servén and Abate (2020), Krisztin and Piribauer (2021), and Shambaugh and Zhou (2024). This literature highlights that macroeconomic and financial outcomes can exhibit systematic dependence across geographically connected economies. Our contribution is to bring this spatial perspective into a model of international portfolio allocation by allowing destination-specific attractiveness to contain a spatially correlated persistent component.

In terms of modeling, our framework is related to dynamic spatial equilibrium models, including Allen and Arkolakis (2025) and Eckert and Kleineberg (2024), as well as the asset demand system developed by Koijen and Yogo (2019, 2026). More specifically, our model is closely related to

Kleinman et al. (2026) and Pellegrino et al. (2025), who embed international asset demand into open-economy general equilibrium models to study the allocation of capital across countries. Pellegrino et al. (2025), in particular, study how bilateral geographic frictions, taxation, and political risk shape global capital allocation. Our paper instead studies how geography shapes international capital allocation through two distinct channels: bilateral investor–destination portfolio frictions and spatially correlated destination-specific heterogeneity. We then quantify the contribution of each channel to cross-country interest-rate dispersion.

2 Data

Interest Rates and Inflation Interest-rate data are obtained from the Global Macro Database. To measure returns on capital while minimizing the influence of short-run monetary policy movements, we use long-term interest rates. For countries with incomplete coverage in the Global Macro Database, including China, Indonesia, Peru, and the Philippines, we supplement the series with 10-year government bond yields from Covered Interest Rate Parity Deviations Between Government Bonds by Wenxin Du. Real interest rates are constructed using five-year-ahead inflation expectations from the IMF World Economic Outlook Historical Forecasts.

Bilateral Capital Holdings We construct bilateral portfolio holdings as the sum of short- and long-term debt securities and equity securities. Bilateral holdings are obtained from the IMF Coordinated Portfolio Investment Survey (CPIS). To construct total amounts outstanding, we combine information from the OECD, BIS Debt Securities Statistics, and World Bank market capitalization data. For the United States, we supplement the CPIS data with Treasury International Capital (TIC) data. To address distortions arising from financial centers and tax havens, we reallocate portfolio positions from a residency basis to an ultimate nationality basis using the Global Capital Allocation Project of Coppola et al. (2021). The implementation follows the data construction in the replication package of Koijen and Yogo (2026).

Country Characteristics We compile country-level and bilateral characteristics from several sources. Country risk, measured by credit default swap spreads, is obtained from Refinitiv Datastream. Capital-account openness is measured using the Chinn–Ito index, and political stability is obtained

from the World Bank Worldwide Governance Indicators. To capture geopolitical alignment, we use United Nations General Assembly ideal-point estimates from Bailey, Strezhnev, and Voeten and geopolitical bloc classifications from the replication package of Gopinath et al. (2025). Bilateral geographic distance and other country-pair characteristics are obtained from CEPII.

3 Motivating Empirical Evidence

This section investigates the spatial relationship in interest rates across countries. If frictions to international capital allocation have a geographic dimension, or if factors relevant for interest rates are spatially correlated, interest rates should exhibit a systematic geographic pattern across countries. In particular, countries that are geographically closer may display more similar interest rates, whereas interest-rate differentials may increase with bilateral distance. To examine this relationship, I consider two empirical specifications. The first exploits time variation in bilateral interest-rate differentials, while the second isolates their time-invariant, persistent component.

3.1 Geographic Gradient in Interest-Rate Differentials

We consider following regression.

$$|r_{i,t} - r_{j,t}| = \beta \ln(dist_{i,j}) + X_{i,j}\Gamma + \alpha_i + \alpha_j + \varepsilon_{ijt}$$

For each country pair, the interest-rate differential is defined as absolute value of rate differences. Interest rates are measured in basis points, so that a one-unit difference in the dependent variable corresponds to a one-percentage-point interest-rate gap. The vector $X_{i,j}$ contains bilateral controls, including indicators for whether the two countries share a border or a common language, have a colonial relationship, or were historically part of the same country. The specification also includes fixed effects for the origin- and destination countries, α_i and α_j, respectively, as well as year fixed effects λ_t. The coefficient of interest, β, measures how the interest-rate gap varies with geographic distance. A positive value of β indicates that the interest-rate differential tends to widen as the geographic distance between the two countries increases. The high- and low-country fixed effects absorb systematic differences associated with each country depending on its position in the interest-rate ranking. Intuitively, the country fixed effects allow us to compare how the interest-rate gap

varies across partners of a given country. For example, conditional on the fixed effect associated with the United States, the regression compares whether the U.S. interest-rate gap tends to be larger with geographically more distant countries.

Panel A in Table 1 shows that geographically more distant country pairs tend to exhibit larger interest-rate differentials. An estimate of $\beta = 0.428$ implies that doubling bilateral distance is associated with an approximately 0.4-basis-point increase in the interest-rate gap. The coefficient remains similar in magnitude after controlling for bilateral relationships, including shared borders, common language, colonial ties, and historical state relationships, suggesting that the geographic gradient is not driven by these bilateral characteristics alone.

3.2 The Persistent Component of Interest-Rate Differentials

The baseline specification establishes that interest-rate differentials are larger between geographically distant countries. However, this relationship may partly reflect transitory country-specific shocks. For example, geographically proximate countries may experience similar monetary or macroeconomic shocks in a given year, generating temporarily similar interest rates. I therefore examine whether geographic distance is also related to the persistent component of bilateral interest-rate differentials. I first decompose the bilateral interest-rate differential into origin-country-by-year and destination-country-by-year components and a time-invariant country-pair component, $\delta_{i,j}$. The country-by-year fixed effects absorb time-varying shocks specific to either member of the pair. The estimated pair component therefore captures the portion of the bilateral interest-rate differential that persists over time after removing these country-specific fluctuations.

$$|r_{i,t} - r_{j,t}| = \alpha_{i,t} + \alpha_{j,t} + \underbrace{\delta_{i,j}}_{\text{pair FE}} + u_{i,j,t}$$

In the second stage, I regress the estimated pair component on bilateral geographic distance and the same set of bilateral controls used in the baseline specification.

$$\hat{\delta}_{i,j} = \beta^P \ln(dist_{i,j}) + X_{i,j}\Gamma + \eta_{i,j}$$

Panel B in Table 1 reports the regression estimates, while Figure 1 (b) presents the corresponding binscatter of the persistent interest-rate differential against log bilateral distance. Both show a strong

positive geographic gradient. The baseline estimate of β^P implies that bilateral distance is associated with a larger persistent interest-rate gap. The coefficient remains similar across specifications that control for bilateral relationships.

Table 1: Geographic Distance and Interest-Rate Differentials

Panel A. Pooled Panel

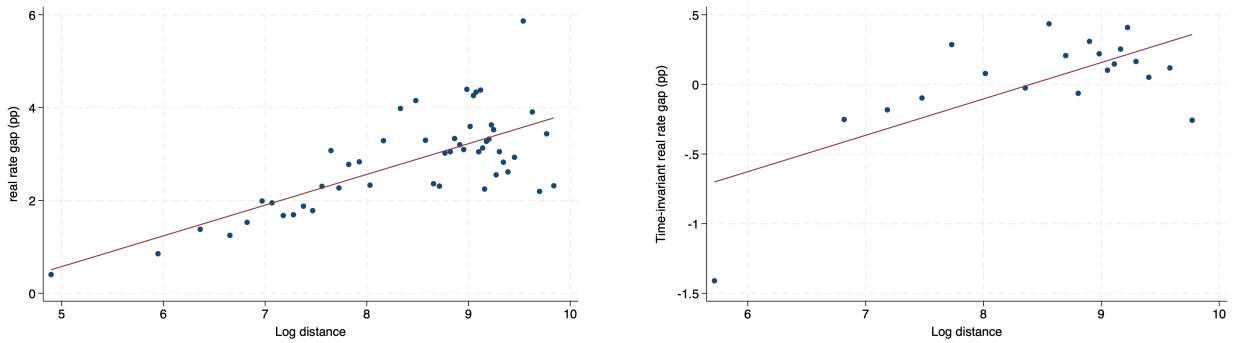
	(1)	(2)	(3)	(4)
Log distance	0.428***	0.458***	0.440***	0.470***
	(0.069)	(0.080)	(0.071)	(0.082)
Shares border		0.364**		0.220
		(0.167)		(0.157)
Common language			0.245**	0.182
			(0.114)	(0.119)
Colonial relationship				0.151
				(0.115)
Same country (past/now)				0.368*
				(0.190)
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	57100	57100	57100	57100
Adj. R-squared	0.487	0.487	0.487	0.487

Panel B. Persistent Pair Component

	(1)	(2)	(3)	(4)
Log distance	0.261***	0.275***	0.261***	0.283***
	(0.023)	(0.025)	(0.023)	(0.025)
Shares border		0.226		0.101
		(0.146)		(0.155)
Common language			-0.004	-0.082
			(0.079)	(0.082)
Colonial relationship				0.447***
				(0.158)
Same country (past/now)				0.354
				(0.244)
Observations	4323	4323	4323	4323
Adj. R-squared	0.028	0.029	0.028	0.030

4 Model

In this section, we develop a dynamic spatial production economy with multiple countries. There are N countries, indexed by $n \in \{1, \ldots, N\}$. Each country is populated by a representative household and competitive firms. Households choose consumption and saving, while their savings are allocated



(a) Pooled panel (b) Persistent component

Figure 1: Interest-Rate Differentials and Geographic Distance

across domestic and foreign investment opportunities through financial intermediaries. International portfolio allocation is subject to bilateral frictions, so that otherwise identical investment opportunities may be valued differently across investor–destination pairs. This setup is consistent with the recent literature on segmented international capital markets, following Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021, 2025), and is closely related to the international portfolio-allocation frameworks in Pellegrino et al. (2025) and Kleinman et.al. (2026).

4.1 Households

The representative household in country n is endowed with a fixed amount of labor, ℓ_n. The household chooses consumption and saving to maximize lifetime utility,

$$\max_{\{c_{nt},a_{n,t+1}\}} \sum_{t=0}^{\infty} \beta_n^t \, u(c_{nt}) \tag{1}$$

$$\text{subject to } c_{n,t} + a_{n,t+1} = \mathcal{R}_{n,t} a_{n,t} + w_{n,t}\ell_n$$

where β_n is the country-specific discount factor,[1] and $\mathcal{R}_{n,t}$ denotes the return on the household's savings portfolio. The optimal consumption-saving decision satisfies $u'(c_{n,t}) = \beta_n \mathcal{R}_{n,t+1} u'(c_{n,t+1})$. In steady state, consumption and portfolio returns are constant over time, so the Euler equation reduces to $1 = \beta_n \mathcal{R}_n$.

[1] Cross-country differences in discount factors may reflect differences in mortality risk, as in Pellegrino et al. (2025), or differences in time preferences, as documented by Falk et al. (2018).

4.2 International Capital Allocation

An investor in country n considering investment in country i faces a bilateral portfolio wedge Φ_{ni}. The wedge captures frictions that reduce the relative attractiveness of investing in a particular foreign destination. We allow the wedge to depend on both geographic distance and geopolitical alignment. First, we assume that the wedge increases with bilateral geographic distance, reflecting informational frictions, limited familiarity, or other barriers associated with investing in more distant markets. Second, motivated by the recent literature on geoeconomic fragmentation, which documents that cross-border FDI and portfolio holdings tend to be lower between geopolitically misaligned countries (e.g., Gopinath et al., 2025), we allow the wedge to increase when the investor and destination belong to different geopolitical blocs. Thus, Φ_{ni} captures both geographic and geopolitical barriers to international capital allocation.

The term ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. Unlike the bilateral portfolio wedge Φ_{ni}, which varies across investor–destination pairs, ω_{it} depends only on the destination country i and time t. It therefore captures characteristics of country i that affect its attractiveness as an investment destination uniformly across investors, beyond the observed interest rate r_{it}. A higher ω_{it} increases the portfolio share allocated to country i by investors from all countries. This distinction is important for the analysis below. While Φ_{ni} captures bilateral segmentation between investor n and destination i, ω_{it} captures destination-side heterogeneity that is common across investors.

For each unit of savings, the intermediary draws an idiosyncratic taste shock ε_{ni} for each destination i, independently across destinations, from a Gumbel distribution.

$$F(x) = \exp(-\exp(-x + \bar{\gamma}))$$

where $\bar{\gamma}$ is Euler–Mascheroni constant. The scale of the idiosyncratic shock is governed by σ_N. The intermediary in country n invests in country i if the friction- and preference-adjusted return from investing in i is the highest among all available destinations.

$$\varepsilon(n,i,\nu)\sigma_N - \Phi_{ni} + \omega_{it} + r_{i,t} = \max_h \{\varepsilon(n,i,\nu)\sigma_N - \Phi_{nh} + \omega_{ht} + r_{h,t}\} \tag{2}$$

The share of country n's savings allocated to country i through financial intermediaries is given by

$$b_{nit} = \frac{a_{nit}}{a_{nt}} = \frac{\exp(-\Phi_{ni} + \omega_i + r_{i,t})^{1/\sigma_N}}{\sum_h \exp(-\Phi_{nh} + \omega_h + r_{h,t})^{1/\sigma_N}} \tag{3}$$

Here, a_{nit} denotes the amount of country n's savings allocated to country i, while a_{nt} denotes total savings in country n. A larger portfolio wedge Φ_{ni} reduces the share allocated to destination i, whereas a higher destination-specific attractiveness ω_{it} or a higher interest rate r_{it} increases its portfolio share. The parameter σ_N governs the dispersion of idiosyncratic investment preferences and, therefore, the sensitivity of portfolio allocation to differences in returns and destination characteristics. The gross return on the household's savings portfolio is then the portfolio-share-weighted average of destination returns:

$$\mathcal{R}_{n,t} = 1 + \sum_i b_{nit} r_{i,t} \tag{4}$$

Country Creditworthiness. One potential source of destination attractiveness is country creditworthiness. Countries with greater creditworthiness may offer more attractive investment opportunities, holding observed returns fixed. To formalize this mechanism, consider the decomposition

$$\omega_{it} = \gamma C_{it} + \chi_{it}, \tag{5}$$

where C_{it} measures country creditworthiness, with higher values indicating greater creditworthiness, and χ_{it} captures other destination-specific characteristics. The parameter $\gamma > 0$ represents the hypothesized sensitivity of investment attractiveness to creditworthiness.

This decomposition does not alter the portfolio allocation equation. It provides an economic interpretation of destination attractiveness, which is recovered without imposing equation (5).

4.3 Firm

Each country n is populated by a representative firm operating a Cobb–Douglas production technology. The firm takes the wage $w_{n,t}$ and the interest rate $r_{n,t}$ as given and chooses its capital stock to maximize the present value of profits. Output is produced using capital $k_{n,t}$ and labor $\ell_{n,t}$, with country-specific productivity $z_{n,t}$. Capital depreciates at rate δ, so that investment required to carry capital into the next period is $k_{n,t+1} - (1-\delta)k_{n,t}$.

The optimality condition for capital implies that the equilibrium return to capital is equal to its

marginal product net of depreciation,

$$r_{n,t+1} = \alpha z_{n,t} k_{n,t}^{\alpha-1} \ell_{n,t}^{1-\alpha}$$

Thus, the interest rate in each country is decreasing in the amount of capital allocated to that country. An increase in capital inflows raises $k_{n,t}$, lowers the marginal product of capital, and thereby reduces the equilibrium interest rate. This relationship provides the link between international portfolio allocation and cross-country interest-rate differences in the model.

4.4 Market Equilibrium

A competitive equilibrium consists of allocations $\{a_{nt}, b_{nit}, k_{nt}\}_{n,i,t}$ and prices $\{\mathcal{R}_{nt}, r_{nt}\}_{n,t}$ such that households optimize given prices and, for all n and t,

$$r_{nt} = \alpha z_n k_{nt}^{\alpha-1}, \tag{6}$$

$$k_{nt} = \sum_i b_{int} a_{it}, \tag{7}$$

$$\mathcal{R}_{nt} = 1 + \sum_i b_{nit} r_{it}. \tag{8}$$

4.5 Destination Attractiveness and Equilibrium Interest Rates

We examine how destination attractiveness affects equilibrium interest rates through international capital allocation. Consider an equilibrium in which investor asset stocks $\{A_n\}$ are fixed, while portfolio shares, destination capital stocks, and interest rates adjust endogenously.

Proposition 1 (Destination Attractiveness and Interest Rates)**.** Suppose the equilibrium is interior, with strictly positive portfolio shares and decreasing marginal products of capital. For any two distinct destinations i and j,

$$-1 < \frac{\partial r_i}{\partial \omega_i} < 0, \qquad \frac{\partial (r_i - r_j)}{\partial \omega_i} < 0. \tag{9}$$

The proof is provided in Appendix B.1.

An increase in destination attractiveness raises investment demand, inducing capital inflows that increase the destination's capital stock and lower its marginal product of capital. The resulting decline in the equilibrium interest rate partially offsets the initial increase in attractiveness. Capital

reallocation also affects returns in other destinations, generating changes in bilateral interest-rate differentials.

The proposition also provides a mechanism through which country-specific characteristics may affect equilibrium interest rates. For example, suppose destination attractiveness depends positively on country creditworthiness, $\omega_i = \gamma C_i + \chi_i$, where $\gamma > 0$. Holding other economic primitives fixed, we obtain

$$\frac{\partial r_i}{\partial C_i} = \gamma \frac{\partial r_i}{\partial \omega_i} < 0. \tag{10}$$

Thus, greater creditworthiness may lower equilibrium interest rates by increasing investment demand and inducing capital inflows.

Consider an equilibrium with fixed investor asset stocks $\{A_n\}$, allowing portfolio shares, capital stocks, and equilibrium interest rates to adjust endogenously.

Proposition 2 (Destination Attractiveness and Interest-Rate Differentials)**.** Consider an interior N-country equilibrium with decreasing marginal products of capital and strictly positive portfolio shares.

For any two distinct destinations i and j, an increase in their relative attractiveness, holding their average attractiveness and all other destination characteristics fixed, reduces their equilibrium interest-rate differential:

$$\frac{\partial (r_i - r_j)}{\partial (\omega_i - \omega_j)} = -\frac{1}{2} \left(P_{ii} + P_{jj} - P_{ij} - P_{ji} \right) < 0, \tag{11}$$

where

$$P = (H^{-1} + J)^{-1} J \tag{12}$$

is the equilibrium interest-rate transmission matrix.

Furthermore, if the equilibrium is symmetric across destinations, the first-order response simplifies to

$$d(r_i - r_j) = -\kappa \, d(\omega_i - \omega_j), \qquad 0 < \kappa < 1. \tag{13}$$

The proof is provided in Appendix B.2.

Proposition 2 establishes that greater relative destination attractiveness reduces the corresponding equilibrium interest-rate differential. Higher attractiveness induces capital inflows, increasing the destination's capital stock and lowering its marginal product of capital.

In the general multi-country economy, this adjustment depends on the entire portfolio allocation structure. Under symmetry, however, the relative interest-rate response is proportional to the difference in destination attractiveness, with a common transmission coefficient κ.

Proposition 3 (Spatial Attractiveness and Interest-Rate Differentials)**.** Consider the symmetric N-country equilibrium in Proposition 2. Let u_i denote a perturbation to destination attractiveness around the symmetric equilibrium, and let $\dot{r}_i$ denote the corresponding first-order equilibrium interest-rate response.

For any two destinations i and j,

$$\dot{r}_i - \dot{r}_j = -\kappa(u_i - u_j), \qquad 0 < \kappa < 1. \tag{14}$$

Consequently, the variance of equilibrium interest-rate differentials satisfies

$$\operatorname{Var}(\dot{r}_i - \dot{r}_j \mid d_{ij}) = \kappa^2 \operatorname{Var}(u_i - u_j \mid d_{ij}). \tag{15}$$

If the variance of bilateral differences in destination attractiveness is increasing in geographic distance,

$$\frac{\partial \operatorname{Var}(u_i - u_j \mid d_{ij})}{\partial \ln d_{ij}} > 0, \tag{16}$$

then

$$\frac{\partial \operatorname{Var}(\dot{r}_i - \dot{r}_j \mid d_{ij})}{\partial \ln d_{ij}} > 0. \tag{17}$$

Moreover, the expected absolute interest-rate differential satisfies

$$\mathbb{E}[|\dot{r}_i - \dot{r}_j| \mid d_{ij}] = \kappa\, \mathbb{E}[|u_i - u_j| \mid d_{ij}]. \tag{18}$$

Thus, an increasing expected absolute difference in destination attractiveness also implies an increasing expected absolute interest-rate differential.

The proof is provided in Appendix B.3.

Proposition 3 establishes that geographic differences in destination attractiveness translate into geographic differences in equilibrium interest rates. In a symmetric multi-country economy, the transmission coefficient is common across destinations, so the geographic gradient in interest-rate differentials inherits the corresponding gradient in destination attractiveness.

Importantly, this result does not require a particular distribution or spatial covariance function. In

the quantitative model, heterogeneous production structures and bilateral portfolio frictions may alter the transmission of spatial attractiveness into equilibrium interest rates.

Corollary 1 (Spatial Creditworthiness)**.** Under the conditions of Proposition 3, suppose destination attractiveness satisfies $\omega_i = \gamma C_i + \chi$, where $\gamma > 0$ and χ is common across destinations.
If the variance of bilateral creditworthiness differences is increasing in geographic distance, then

$$\frac{\partial \operatorname{Var}(\dot{r}_i - \dot{r}_j \mid d_{ij})}{\partial \ln d_{ij}} > 0. \tag{19}$$

Similarly, an increasing expected absolute creditworthiness difference implies an increasing expected absolute interest-rate differential.

5 Parameterization

5.1 Portfolio Friction Estimation

This section estimates bilateral portfolio frictions, $\phi_{ni} = \Phi_{ni}/\sigma_N$, using PPML. From the portfolio allocation equation (3), investment shares can be decomposed into three components. The first is the bilateral portfolio friction, ϕ_{ni}, which varies across investor–destination pairs and captures barriers specific to international capital allocation. We parameterize this component as a function of bilateral geographic distance, home bias, and geopolitical alignment:

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta^{foreign} \mathbb{1}[n \neq i] + \beta^{crossblock} Crossblock$$

The coefficient β^{dist} captures the geographic component of bilateral portfolio frictions. The coefficient $\beta^{foreign}$ captures home bias, or the additional friction associated with investing abroad relative to investing domestically. Finally, $\beta^{crossblock}$ captures the additional friction associated with investing in a destination that belongs to a different geopolitical bloc. Countries are classified into three groups—the U.S. bloc, the China bloc, and non-aligned countries—based on their ideal points in the United Nations General Assembly.
Second, the destination-specific component, $(\omega_{it}+r_{it})/\sigma_N$, is common across investors and is absorbed by destination-country $\times$ time fixed effects. Third, the denominator of the allocation equation captures investor-specific aggregate demand for foreign assets and is absorbed by investor-country $\times$ time fixed effects.

$$b_{nit} = \underbrace{\frac{\overbrace{\exp(-\phi_{ni})}^{\text{Capital Moving Friction}} \overbrace{\exp(\omega_{it}/\sigma_N + r_{it}/\sigma_N)}^{\text{Time}\times\text{Destin FE}}}{\sum_h \exp(-\phi_{nh} + \omega_{ht}/\sigma_N + r_{ht}/\sigma_N))}}_{\text{Time}\times\text{Origin FE}} \tag{20}$$

Estimation Results Table 2 reports the PPML estimates of bilateral portfolio frictions. The coefficient on log distance is negative and statistically significant, indicating that geographic distance reduces bilateral portfolio allocation. The estimate of $\beta_d = -0.797$ implies that a 1 percent increase in bilateral distance is associated with a 0.854 percent decline in bilateral portfolio holdings. In terms of economic magnitude, doubling bilateral distance is associated with an approximately 42 percent reduction in predicted bilateral portfolio holdings.

The large negative coefficient on the foreign indicator captures substantial home bias in international portfolio allocation. The negative coefficient on the cross-bloc indicator shows that, conditional on geographic distance and home bias, bilateral portfolio holdings are lower when the investor and destination countries belong to different geopolitical blocs.[2] These results imply that geographic distance represents an important bilateral portfolio friction in international capital allocation.

[2]The magnitude is broadly comparable to recent estimates of financial fragmentation using continuous measures of geopolitical distance. For example, recent gravity estimates find that a one-standard-deviation increase in geopolitical distance is associated with roughly a 40 percent decline in equity allocations and a 60 percent decline in bond allocations by investment funds. These estimates are not directly comparable to ours because our specification uses a discrete cross-bloc indicator.

Table 2: Bilateral Portfolio Frictions and Geographic Distance

	(1)
	share
$\ln(dist_{i,j})$	-0.797***
	(0.0544)
Foreign	-4.159***
	(0.130)
Cross Block	-0.796***
	(0.155)
Observations	100377

Standard errors in parentheses

Standard errors clustered by country-pair (country x counterpart).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2 σ_N Estimation

The return term is endogenous because equilibrium interest rates are jointly determined with international portfolio allocation. To estimate σ_N, we use an IV strategy that builds on Koijen and Yogo (2026) and adapts their residual-supply approach to our international capital-allocation setting. The instrument exploits variation in predicted asset demand relative to predicted asset supply. This residual demand pressure shifts equilibrium returns through asset-market clearing but, under the exclusion restriction, does not directly affect investor n's bilateral portfolio demand. It therefore provides exogenous variation in returns that identifies the sensitivity of portfolio allocation to equilibrium returns.

Equation (3) can be modified to be expressed as

$$\log \frac{b_{ni}}{b_{nn}} = \frac{1}{\sigma_N}(r_{it} - r_{nt} - \Phi_{ni} + \omega_{it} - \omega_{nt})$$

Let X_{ni} denote the observed bilateral characteristics between investor country n and destination country i, including bilateral geographic distance and the cross-bloc indicator. The identification

assumptions for the IV strategy are:

$$\text{Exogeneity:} \quad \text{Cov}\left(\varepsilon_{nit},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) = 0,$$

$$\text{Relevance:} \quad \text{Cov}\left(r_{it},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) \neq 0, \qquad j \neq n.$$

The exogeneity condition requires that, conditional on the bilateral characteristics between investor country n and destination country i, the geographic distance between other investor countries j and destination i is uncorrelated with the idiosyncratic portfolio-demand shock of investor n. Intuitively, the location of other investor countries should not directly affect country n's relative preference for destination i.

The relevance condition requires that the geographic distribution of other investor countries affects the equilibrium return in destination i. Because bilateral portfolio demand declines with geographic distance, destinations that are relatively close to countries with greater investment capacity face stronger predicted demand. Through capital-market clearing, this variation in demand pressure affects the equilibrium return r_{it}. We aggregate this cross-country distance variation into the residual-demand instrument constructed below.

Demand Component From equation (3), we construct the time-invariant component of portfolio demand using only the estimated bilateral gravity frictions. Specifically, the predicted allocation share from investor country m to destination country i is given by

$$\hat{b}_{mi}^{gravity} = \frac{\exp(-\phi_{mi}^{gravity})}{\sum_h \exp(-\phi_{mh}^{gravity})},$$

where

$$\phi_{ni}^{gravity} = \hat{\beta}^{dist} \ln(dist_{ni}) + \hat{\beta}^{foreign} \mathbb{1}[n \neq i].$$

This predicted share captures the time-invariant geographic structure of portfolio demand implied by bilateral distance and home bias, abstracting from time-varying destination characteristics and equilibrium returns. We then aggregate predicted demand from all investor countries other than n:

$$\sum_{m \neq n} A_{mt} \hat{b}_{mi}^{gravity}.$$

Supply Component To construct the supply component, we estimate the following panel regression:

$$\log S_{it} = \Psi' z_{it} + \lambda_t + \varepsilon_{it}^S,$$

where $z_{it} = [\log GDP_{it}, \log pop_{it}]'$, and λ_t denotes year fixed effects. We then construct predicted asset supply as

$$\log \hat{S}_{it} = \hat{\Psi}' z_{it} + \hat{\lambda}_t.$$

The predicted value captures the component of asset supply explained by observable country characteristics and common time variation.

Residual Demand Pressure Using the predicted asset allocation from the gravity model, we first construct a leave-one-out measure of demand pressure for each destination i:

$$Z_{i,-n,t} = \log \left(\sum_{m \neq n} A_{mt} \hat{b}_{mit}^{gravity} \right) - \log \hat{S}_{it}.$$

This measure is the negative of the conventional residual-supply measure. A larger value of $Z_{i,-n,t}$ therefore indicates stronger predicted demand relative to asset supply in destination country i. The leave-one-out construction excludes predicted demand originating from investor country n, so that the bilateral portfolio choice of investor n does not mechanically enter the demand component. Because the endogenous regressor is the return differential between destination i and country n, we use the corresponding difference in residual demand pressure as the instrument:

$$Z_{nit}^{rel} = Z_{i,-n,t} - Z_{n,-n,t}.$$

Thus, Z_{nit}^{rel} measures the relative predicted demand pressure on destination i compared with country n, generated by investors other than n. Variation in this relative demand pressure shifts the equilibrium return differential $r_{it} - r_{nt}$ through asset-market clearing.

2SLS The return elasticity of portfolio demand is estimated using two-stage least squares. Since the portfolio-demand equation is expressed in terms of returns relative to the investor's domestic asset, we instrument the return differential $r_{it} - r_{nt}$ with the corresponding difference in leave-one-out

residual demand pressure. The first stage is

$$r_{it} - r_{nt} = \pi Z^{rel}_{nit} + \mathbf{X}'_{ni}\beta + \nu_{nit},$$

where

$$Z^{rel}_{nit} = Z_{i,-n,t} - Z_{n,-n,t}.$$

The estimated return coefficient implies $\hat{\sigma}_N = \frac{1}{\hat{\lambda}_r} \approx 0.075$

Table 3: IV Estimation Results

	(1)	(2)
	First stage	2SLS
Relative Z	0.01***	
	(0.00)	
ΔR		13.58***
		(3.39)
First-stage F (excl. instr.)		688.33
Kleibergen-Paap rk Wald F		688.33
Observations	28772	28772
Adjusted R^2		0.092

Fixed effects: country and counterpart. Robust standard errors in parentheses.

5.3 Steady-State Calibration

The remaining country-specific parameters are calibrated to match the baseline steady state. We choose β_n to satisfy the steady-state Euler equation and calibrate z_n to match the observed interest rate. These parameters are held fixed in the counterfactual exercises.

6 Spatial Structure and Economic Origins of Destination Attractiveness

6.1 Geographic Patterns in Destination Attractiveness

In the model, ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. If destination characteristics are spatially correlated, geographically proximate countries should exhibit more similar values of ω_{it}, generating geographic clusters in asset attractiveness. This provides a second channel through which geography affects international capital allocation, distinct from the bilateral portfolio friction ϕ_{ni}. The distinction is that ϕ_{ni} operates between an investor and a destination, whereas the spatial component of ω_{it} operates across destination countries themselves. For example, a U.S. investor may allocate less capital to Korea than to Canada because Korea is geographically more distant from the United States. This bilateral channel, however, does not capture the fact that Korea may share destination characteristics with nearby countries such as Japan. If ω_{it} is spatially correlated, investors from all origins may perceive Korean assets as more similar to Japanese assets than to assets in geographically distant countries such as Argentina. The spatial structure of ω_{it} captures this destination-side geographic dependence. To examine whether destination attractiveness exhibits a geographic pattern, I first relate bilateral differences in ω_{it} to geographic distance:

$$|\omega_{it} - \omega_{jt}| = \beta^{\omega} \ln(dist_{ij}) + X'_{ij}\Gamma + \alpha_i + \alpha_j + \lambda_t + \varepsilon_{ijt}.$$

A positive coefficient on distance indicates that geographically distant countries tend to have more dissimilar destination components. Table 4a shows a positive and statistically significant relationship between bilateral distance and differences in ω_{it}. The relationship is robust to controls for contiguity, common language, colonial ties, and historical state relationships. Part of this relationship may reflect observable destination characteristics that themselves vary systematically across space. In particular, the recovered destination component may contain variation associated with market size. I therefore remove observable destination characteristics and year effects from ω_{it},

$$u_{it} = \omega_{it} - X'_{it}\hat{\Gamma} - \hat{\lambda}_t, \tag{21}$$

where the baseline specification includes log population in X_{it}. I then estimate the analogous bilateral regression

$$|u_{it} - u_{jt}| = \beta^u \ln(dist_{ij}) + Z'_{ij}\Pi + \alpha_i + \alpha_j + \lambda_t + \nu_{ijt}.$$

The positive relationship with geographic distance remains after removing market size component. Moreover, geopolitical distance is also positively associated with differences in the residual destination component. These patterns motivates the spatial decomposition developed below.

(a) Distance Regression on ω differences

	(1)	(2)	(3)	(4)
ln(distance)	0.0232***	0.0252***	0.0245***	0.0262***
	(0.0037)	(0.0041)	(0.0038)	(0.0042)
Contiguity		0.0237***		0.0138
		(0.0090)		(0.0090)
Common language			0.0263***	0.0238***
			(0.0072)	(0.0072)
Colonial link				0.0010
				(0.0105)
Same country				0.0193*
				(0.0102)
Country 1 FE	Yes	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes	Yes
Year FE	YES	YES	YES	YES
Observations	57,100	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324	4,324
Adj. R-squared	0.580	0.581	0.581	0.582

(b) Distance Regression on u differences

	(1)	(2)	(3)
ln(distance)	0.0263***		0.0099***
	(0.0015)		(0.0016)
Ideal-point distance		0.0620***	0.0551***
		(0.0028)	(0.0031)
Country 1 FE	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324
Adj. R-squared	0.042	0.099	0.103

Table 4: Spatial relationships of destination attractiveness

6.2 Spatial Decomposition

To characterize the spatial structure of destination attractiveness, we decompose the residual component u_{it} into a persistent spatial component, s_i, a persistent non-spatial component, a_i, and a

transitory component, ε_{it}:

$$u_{it} = \underbrace{s_i}_{\substack{\text{Spatial} \\ \text{Component}}} + \underbrace{a_i}_{\substack{\text{Non-spatial} \\ \text{Component}}} + \varepsilon_{it}.$$

As in a random-effects framework, we assume that the components are Gaussian

$$s \sim N(0, \Sigma_s), \qquad a \sim N(0, \sigma_a^2 I), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2).$$

To capture spatial dependence, we parameterize the spatial covariance matrix using a Gaussian covariance function [3]

$$[\Sigma_s]_{ij} = \sigma_s^2 \exp\left(-\left(\frac{d_{ij}}{\rho}\right)^2\right)$$

, where σ_s is scale parameter of spatial components while ρ is decay parameter which measures how fast spatial dependence decay over distance.

6.3 Identification and Estimation of Parameters

Let $u_{it} = s_i + a_i + \varepsilon_{it}$ denote the latent deviation from the conditional mean. We assume that the spatial component, the non-spatial persistent component, and the transitory component are mutually orthogonal. We further assume that a_i is uncorrelated across countries and that ε_{it} is uncorrelated across countries and over time. Averaging over time gives

$$\bar{u}_i = \frac{1}{T_i} \sum_t u_{it} = s_i + a_i + \bar{\varepsilon}_i.$$

The identifying information comes from covariance across countries and over time. For two different countries $i \neq j$,

$$\mathrm{Cov}(\bar{u}_i, \bar{u}_j) = \sigma_s^2 \exp\left[-\left(\frac{d_{ij}}{\rho}\right)^2\right],$$

where d_{ij} denotes chordal distance.[4] Under the orthogonality assumptions, cross-country covariance reflects only the spatial component. Its magnitude and variation with bilateral distance provide identifying information about σ_s^2 and ρ. Separate identification requires sufficiently informative variation in distances; identification can be weak when the spatial kernel is close to an identity matrix or a constant matrix.

[3] The Gaussian covariance function, also known as the squared-exponential covariance function, can be obtained as the limiting case of the Matérn covariance family as the smoothness parameter tends to infinity.

[4] Chordal distance is the Euclidean distance between two points on the Earth's surface embedded in three-dimensional space. Using this distance ensures that the Gaussian covariance kernel is positive semidefinite.

For the same country observed in two different periods,

$$\mathrm{Cov}(u_{it}, u_{i\tau}) = \sigma_s^2 + \sigma_a^2, \qquad t \neq \tau.$$

Given the spatial variance, within-country persistence identifies σ_a^2. Finally,

$$\mathrm{Var}(u_{it}) = \sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2.$$

The remaining variance after accounting for the two persistent components identifies σ_ε^2.

We jointly estimate $(\sigma_s^2, \rho, \sigma_a^2, \sigma_\varepsilon^2)$ by restricted maximum likelihood (REML), controlling for log population and year fixed effects. The estimation uses 1,028 country-year observations from 67 countries over 2003–2020. Table 5 reports the estimates. The estimated spatial variance is $\hat{\sigma}_s^2 = 0.01687$, while the non-spatial persistent and transitory variances are $\hat{\sigma}_a^2 = 0.00816$ and $\hat{\sigma}_\varepsilon^2 = 0.00352$, respectively.

The estimated spatial correlation range is $\hat{\rho} = 2{,}929$ km in chordal distance. Under the Gaussian covariance specification, correlation within the spatial component falls to 0.5 at a great-circle distance of approximately 2,454 km and to 0.1 at approximately 4,541 km.

A parametric-bootstrap likelihood-ratio test with 500 replications rejects the null hypothesis $H_0 : \sigma_s^2 = 0$, with an LR statistic of 36.609 and a bootstrap p-value of 0.002. This provides evidence against the absence of a spatial persistent component.

Table 5: Spatial Decomposition of ω

Parameter	Estimate	Interpretation
σ_s^2	0.01687	Spatial persistent variance
ρ	2,929 km	Spatial correlation range (chordal)
σ_a^2	0.00816	Non-spatial persistent variance
σ_ε^2	0.00352	Transitory variance
$\sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2$	0.02855	Total variance of u_{it}

6.4 Recovering the Spatial Component

After estimating the variance components, we recover the country-specific spatial and non-spatial persistent components using the Best Linear Unbiased Predictor (BLUP). This approach is closely related to the empirical Bayes and BLUP methods used in the labor economics literature to recover

latent persistent heterogeneity from noisy observed outcomes, as in Kane and Staiger (2008) and Chetty et al. (2014). We apply the same basic idea here, using the estimated spatial covariance structure to separate the latent spatial component from non-spatial country-specific heterogeneity and transitory noise. We can express $\bar{u}_i$ vector form,

$$\bar{u} = s + a + \bar{\varepsilon}.$$

Because averaging reduces the variance of the transitory component, the covariance matrix of $\bar{u}$ is

$$V_{\bar{u}} = \Sigma_s + \sigma_a^2 I + D_\varepsilon,$$

where

$$D_\varepsilon = \operatorname{diag}\left(\frac{\sigma_\varepsilon^2}{T_1}, \ldots, \frac{\sigma_\varepsilon^2}{T_N}\right).$$

The spatial component is recovered from its conditional expectation given the observed average residuals:

$$\hat{s} = E[s \mid \bar{u}] = \Sigma_s V_{\bar{u}}^{-1} \bar{u}.$$

Similarly, the non-spatial persistent component is recovered as

$$\hat{a} = E[a \mid \bar{u}] = \sigma_a^2 I V_{\bar{u}}^{-1} \bar{u}.$$

The predictor assigns variation in $\bar{u}$ to the spatial component according to the estimated cross-country covariance structure. Variation that is persistent but not explained by spatial covariance is instead attributed to the non-spatial country component.

As a validation exercise, we construct a leave-one-country-out (LOO) estimate of the spatial component. For each country, we exclude all observations for that country and predict its spatial component using the outcomes of the remaining countries and the estimated spatial covariance structure. The LOO estimates are highly correlated with the full-sample BLUP estimates, with a correlation of 0.944. This close correspondence indicates suggests that the estimated spatial component reflects systematic cross-country spatial dependence rather than simply absorbing country-specific persistent variation.

6.5 Country Creditworthiness and Spatial Dependence

We investigate country creditworthiness as a potential economic source of spatial dependence in destination attractiveness. We measure creditworthiness using the sovereign credit-rating score from Refinitiv Datastream, where higher values indicate greater creditworthiness.

Spatial Dependence in Creditworthiness. We decompose country creditworthiness into time effects, persistent spatial and non-spatial components, and transitory variation:

$$C_{it} = \alpha_t^C + s_i^C + a_i^C + \varepsilon_{it}^C. \tag{22}$$

Following the spatial specification for destination attractiveness, we characterize the covariance of the persistent spatial component using a Gaussian kernel:

$$K_C(d_{ij}) \equiv \mathrm{Cov}(s_i^C, s_j^C) = \sigma_C^2 \exp\left[-\left(\frac{d_{ij}}{\rho_C}\right)^2\right]. \tag{23}$$

This covariance structure implies that the variance of bilateral creditworthiness differences increases with geographic distance:

$$\mathrm{Var}(s_i^C - s_j^C) = 2\sigma_C^2\left[1 - \exp\left(-\left(\frac{d_{ij}}{\rho_C}\right)^2\right)\right]. \tag{24}$$

In particular,

$$\frac{\partial \,\mathrm{Var}(s_i^C - s_j^C)}{\partial \ln d_{ij}} > 0 \qquad (d_{ij} > 0).$$

Thus, geographically distant countries exhibit greater dispersion in their persistent creditworthiness components.

Connection to Destination Attractiveness. Recall that destination attractiveness is decomposed into country creditworthiness and other destination characteristics as in equation (5). Allowing both components to exhibit persistent spatial dependence, we obtain

$$s_i^\omega = \gamma s_i^C + s_i^\chi. \tag{25}$$

Let $K_{AB}(d_{ij})$ denote the spatial cross-covariance between components s_i^A and s_j^B, assuming these covariances depend only on geographic distance. The covariance of destination attractiveness can

then be expressed as

$$K_\omega(d) = \gamma^2 K_C(d) + K_\chi(d) + \gamma \left[K_{C\chi}(d) + K_{\chi C}(d)\right]. \tag{26}$$

The corresponding cross-spatial covariance is

$$K_{C\omega}(d) = \gamma K_C(d) + K_{C\chi}(d). \tag{27}$$

These expressions show how spatial dependence in creditworthiness may be associated with spatial dependence in destination attractiveness.

Implications for Interest-Rate Differentials. Proposition 1 establishes that destination attractiveness affects equilibrium interest rates through international capital reallocation. Spatially correlated creditworthiness therefore provides a potential mechanism linking geographic distance to interest-rate differentials.

6.6 Empirical Evidence on Country Creditworthiness

We first estimate the spatial covariance of sovereign creditworthiness using the same Gaussian covariance specification employed for destination attractiveness.

We find significant persistent spatial dependence in creditworthiness. We then estimate the joint spatial covariance between creditworthiness and destination attractiveness. The estimated cross-spatial covariance is positive in the expanded country sample.

7 Quantitative Exercise

For the quantitative exercise, we compute the steady state of the model. To construct steady-state counterparts of the data, we average country-level variables over the 2003–2020 period. The model requires a complete $N \times N$ matrix of bilateral portfolio holdings across investor and destination countries. Because not all countries report bilateral portfolio positions, the quantitative sample is restricted to 52 countries for which this matrix can be constructed. Destinations outside the quantitative sample are retained through an aggregate Rest-of-the-World outside option.[5]

[5]Mauritius is included in the Rest-of-the-World outside option because of its small population.

7.1 Decomposition of Geographic Channels

We first quantify the contribution of the two geographic channels holding investor-country asset supplies fixed at their baseline levels. Under each counterfactual, portfolio shares are allowed to adjust, reallocating the fixed global stock of assets across destinations and thereby changing destination capital stocks and equilibrium interest rates. Holding asset supplies fixed isolates the effect of geographic forces through the reallocation of a given global stock of capital, abstracting from the endogenous adjustment in saving implied by the household Euler conditions. We measure population-weighted interest-rate dispersion relative to the United States as

$$D_{US}(r) = \sum_{i \neq US} w_i |r_i - r_{US}|,$$

where w_i denotes country i's population weight among non-U.S. countries. We report the decomposition for the full sample and separately for advanced economies (AE) and emerging market economies (EME).[6]

Table 6: Decomposition of U.S.-Relative Interest-Rate Dispersion

Group	Interest-Rate Dispersion			Decomposition of Change		
	Baseline (%)	No Geographic Channels (%)	Change (pp)	Bilateral Friction (pp)	Spatial Attractiveness (pp)	Interaction (pp)
All	1.443	0.762	0.682	0.000	0.627	0.055
AE	0.494	1.175	-0.681	-0.640	-0.426	0.384
EME	1.575	0.704	0.871	0.089	0.773	0.009

Notes: Interest-rate dispersion is the population-weighted mean absolute interest-rate difference relative to the United States[cite: 1]. The baseline includes both geographic channels[cite: 1]. "No Geographic Channels" removes both the bilateral geographic portfolio friction and the spatial component of destination attractiveness[cite: 1]. The final three columns decompose the resulting change in dispersion into the bilateral-friction component, the spatial-attractiveness component, and their interaction[cite: 1].

Table 6 reports the decomposition. For the full sample, baseline interest-rate dispersion relative to the

[6]Advanced economies include Australia, Austria, Belgium, Canada, Switzerland, Germany, Denmark, Spain, Finland, France, the United Kingdom, Greece, Italy, Japan, the Netherlands, Norway, New Zealand, Portugal, and Sweden. Emerging market economies include Bangladesh, Bulgaria, Brazil, Chile, China, Colombia, Costa Rica, Cyprus, Czechia, Estonia, Hungary, Indonesia, India, Iceland, Israel, the Republic of Korea, Lithuania, Latvia, Mexico, Malta, Malaysia, Pakistan, Peru, the Philippines, Poland, Romania, the Russian Federation, Slovakia, Slovenia, Thailand, Türkiye, Venezuela, and South Africa.

United States is 1.44 percent, compared with 0.76 percent when both geographic channels are removed. The resulting 0.68 percentage-point contribution is driven primarily by the spatial component of destination attractiveness, while the main contribution of bilateral geographic frictions is close to zero in the aggregate. The interaction between the two channels accounts for the remaining increase in dispersion. The aggregate result masks substantial heterogeneity across country groups. Among advanced economies, the presence of the two geographic channels reduces interest-rate dispersion relative to the United States. Both the bilateral-friction and spatial-attractiveness components contribute negatively to dispersion, although their positive interaction partially offsets these effects. In contrast, geographic forces increase interest-rate dispersion among emerging market economies. This increase is driven predominantly by the spatial component of destination attractiveness, with a smaller contribution from bilateral geographic frictions.

7.2 Steady State Counterfactuals

We next allow capital stocks to adjust endogenously and solve for the steady-state equilibrium under the same four scenarios. Unlike the decomposition above, this exercise allows capital allocation and equilibrium interest rates to adjust in response to changes in the geographic channels. Table 7 reports the counterfactual results. Removing bilateral distance frictions reduces the cross-country variance of log capital by 25.0 percent relative to the baseline and the variance of log GDP per capita by 10.8 percent. Removing the spatial component of destination attractiveness also reduces dispersion, lowering the variance of log capital by 28.6 percent and that of log GDP per capita by 9.9 percent. When both geographic channels are removed, the reductions are substantially larger: the variance of log capital falls by 63.8 percent and the variance of log GDP per capita by 24.9 percent. These results indicate that both bilateral portfolio frictions and spatially correlated destination attractiveness contribute to cross-country differences in capital allocation and output in general equilibrium.

Table 7: Counterfactual Dispersion across Scenarios

	Var($\log k$)				Var($\log$ GDP p.c.)			
	Simple		Pop-w		Simple		Pop-w	
Scenario	Level	Δ (%)	Level	Δ (%)	Level	Δ (%)	Level	Δ (%)
Baseline	4.385	–	2.929	–	4.632	–	2.523	–
No distance friction	3.290	−25.0	2.609	−10.9	4.134	−10.8	2.370	−6.1
No spatial component	3.131	−28.6	1.662	−43.3	4.173	−9.9	2.078	−17.6
Neither channel	1.588	−63.8	0.886	−69.7	3.479	−24.9	1.640	−35.0

8 Conclusion

This paper studies how geography shapes international capital allocation and cross-country interest-rate differences. We first document a robust spatial gradient in interest rates. Geographically distant countries exhibit larger interest-rate differentials, and this relationship is robust in the persistent component of interest-rate gaps. These patterns suggest that geography remains relevant for international financial allocation even beyond short-run country-specific fluctuations.

To quantify the mechanisms behind this relationship, we develop a multi-country portfolio-allocation model with two distinct geographic channels. The first is a bilateral portfolio friction that depends on the relationship between an investor country and an investment destination. The second operates through destination-specific attractiveness, which may itself be spatially correlated across countries. We estimate bilateral portfolio frictions from international portfolio holdings. We then decompose destination attractiveness into spatially correlated persistent, non-spatial persistent, and transitory components. The estimated spatial component accounts for a substantial share of persistent cross-country heterogeneity in destination attractiveness.

Holding capital stocks fixed, the decomposition shows that the increase in aggregate interest-rate dispersion associated with geography is driven primarily by the spatial component of destination attractiveness. The aggregate contribution of bilateral geographic frictions is close to zero, but this masks substantial heterogeneity across country groups. Geographic forces reduce interest-rate dispersion relative to the United States among advanced economies, while they increase dispersion

among emerging market economies, with the latter effect driven predominantly by the spatial destination component. When capital stocks are allowed to adjust endogenously, both geographic channels have substantial effects on the cross-country allocation of capital and output. Removing either bilateral distance frictions or the spatial component of destination attractiveness reduces the dispersion of capital and output, and removing both channels produces substantially larger reductions.

Overall, the results show that geography shapes international capital allocation through both bilateral investor–destination frictions and spatial dependence across destination countries. Their quantitative importance differs across outcomes and country groups. Spatial destination characteristics account for most of the aggregate geographic pattern in interest-rate dispersion, while both channels play important roles in determining the distribution of capital and output in general equilibrium.

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Appendix

A Model Solution Algorithm

A.1 Solution Algorithm

We solve the stationary general equilibrium using a nested numerical procedure. All parameters, including the calibrated discount factors $\{\beta_n\}$, are held fixed throughout the solution. We use the normalized bilateral portfolio wedge

$$\phi_{ni} \equiv \frac{\Phi_{ni}}{\sigma_N},$$

as defined in Section 5.1.

For a given vector of asset supplies $\{A_n\}$, the inner loop solves for capital allocation, portfolio shares, and interest rates. Given a current guess for capital $\{k_i\}$, interest rates are

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1} \tag{28}$$

Portfolio shares are then determined by

$$b_{ni} = \frac{\exp\left(-\phi_{ni} + \omega_i/\sigma_N + r_i/\sigma_N\right)}{\sum_j \exp\left(-\phi_{nj} + \omega_j/\sigma_N + r_j/\sigma_N\right)}, \tag{29}$$

with the modification for destinations outside the model sample described in Appendix A.2. The implied capital stock is

$$k_i' = \sum_n b_{ni} A_n. \tag{30}$$

We iterate over $\{k_i\}$ until the implied capital allocation converges.

Given the resulting portfolio allocation and interest rates, the outer loop adjusts asset supplies until the steady-state Euler condition

$$\beta_n R_n = 1 \tag{31}$$

is satisfied for every investor country, where

$$R_n = 1 + \sum_i b_{ni} r_i. \tag{32}$$

Algorithm 1 Solution of the Stationary General Equilibrium

Initialize asset supplies $\{A_n\}$.

repeat

  Initialize destination capital stocks $\{k_i\}$.

  repeat

    Compute interest rates:

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1}$$

    Compute bilateral portfolio shares $\{b_{ni}\}$.

    Update destination capital:

$$k_i' = \sum_n b_{ni} A_n.$$

  until $\{k_i\}$ converges

  Compute investor portfolio returns:

$$R_n = 1 + \sum_i b_{ni} r_i.$$

  Update $\{A_n\}$ to satisfy

$$\beta_n R_n = 1 \qquad \forall n.$$

until the Euler conditions converge

return $\{A_n, b_{ni}, k_i, r_i\}$.

A.2 Treatment of the Rest of the World

The set of destinations observed in the bilateral portfolio data is larger than the set of countries explicitly included in the general-equilibrium model. Dropping these destinations would mechanically reallocate their portfolio shares toward countries in the model sample. We therefore aggregate all destinations outside the model sample into a single outside option.

Let $\mathcal{N}$ denote the set of model countries and $\mathcal{R}$ the set of observed destinations outside $\mathcal{N}$. For

investor n, we construct the inclusive value

$$IV_{nR} = \log \sum_{j \in \mathcal{R}} \exp\left(-\phi_{nj} + v_j\right), \tag{33}$$

where v_j is the estimated destination attractiveness component. The portfolio problem then treats IV_{nR} as an additional investment alternative:

$$b_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in \mathcal{N}} \exp(V_{nj}) + \exp(IV_{nR})}, \qquad V_{ni} = -\phi_{ni} + \frac{\omega_i + r_i}{\sigma_N}. \tag{34}$$

Assets allocated to the outside option do not enter the capital stock of countries in $\mathcal{N}$. Hence, capital-market clearing within the model is

$$k_i = \sum_{n \in \mathcal{N}} b_{ni} A_n, \qquad i \in \mathcal{N}. \tag{35}$$

Because the model does not solve for an endogenous return outside the model sample, portfolio returns entering the Euler equation are computed using returns on model destinations only.

B Proofs of Interest-Rate Comparative Statics

B.1 Proof of Proposition 1

Suppress time subscripts and consider a fixed vector of investor-country asset stocks $\boldsymbol{A}$.

The portfolio share of investor n allocated to destination i is

$$b_{ni} = \frac{\exp[(r_i + \omega_i - \Phi_{ni})/\sigma_N]}{\sum_j \exp[(r_j + \omega_j - \Phi_{nj})/\sigma_N]}. \tag{36}$$

Define the destination-specific attractiveness-adjusted return

$$q_i = r_i + \omega_i. \tag{37}$$

The capital supply to destination i is

$$K_i^s(\boldsymbol{q}) = \sum_n A_n b_{ni}(\boldsymbol{q}). \tag{38}$$

Differentiating the portfolio share with respect to q_j gives

$$\frac{\partial b_{ni}}{\partial q_j} = \frac{1}{\sigma_N} b_{ni} \left(\mathbf{1}\{i = j\} - b_{nj}\right). \tag{39}$$

Define the Jacobian matrix

$$J_{ij} \equiv \frac{\partial K_i^s}{\partial q_j}. \tag{40}$$

Then

$$J_{ij} = \frac{1}{\sigma_N} \sum_n A_n b_{ni} \left(\mathbf{1}\{i = j\} - b_{nj} \right). \tag{41}$$

Equivalently,

$$J = \frac{1}{\sigma_N} \sum_n A_n \left[\mathrm{diag}(\boldsymbol{b}_n) - \boldsymbol{b}_n \boldsymbol{b}_n' \right]. \tag{42}$$

The matrix J is symmetric and positive semidefinite. For strictly positive portfolio shares,

$$J_{ii} > 0, \qquad J_{ij} < 0 \quad (i \neq j). \tag{43}$$

On the production side, the interest rate satisfies

$$r_i = \alpha z_i \left(\frac{k_i}{\ell_i} \right)^{\alpha - 1}. \tag{44}$$

Define

$$h_i \equiv -\frac{\partial r_i}{\partial k_i} = (1 - \alpha) \frac{r_i}{k_i} > 0 \tag{45}$$

and

$$H = \mathrm{diag}(h_1, \ldots, h_N). \tag{46}$$

The production condition implies

$$\mathrm{d}\boldsymbol{r} = -H \, \mathrm{d}\boldsymbol{k}. \tag{47}$$

Capital market clearing requires

$$\boldsymbol{k} = K^s(\boldsymbol{r} + \boldsymbol{\omega}). \tag{48}$$

Holding investor asset stocks fixed, totally differentiating the market-clearing condition gives

$$\mathrm{d}\boldsymbol{k} = J(\mathrm{d}\boldsymbol{r} + \mathrm{d}\boldsymbol{\omega}). \tag{49}$$

Combining the two equations,

$$\mathrm{d}\boldsymbol{r} = -HJ(\mathrm{d}\boldsymbol{r} + \mathrm{d}\boldsymbol{\omega}). \tag{50}$$

Rearranging yields

$$(H^{-1} + J)\mathrm{d}\boldsymbol{r} = -J \, \mathrm{d}\boldsymbol{\omega}. \tag{51}$$

Therefore, the general-equilibrium interest-rate response is

$$\frac{\partial \boldsymbol{r}}{\partial \boldsymbol{\omega}'} = -(H^{-1} + J)^{-1} J. \tag{52}$$

Let

$$M = H^{-1} + J. \tag{53}$$

Since H^{-1} is positive definite and J is positive semidefinite, M is positive definite.

Using

$$M^{-1} J = I - M^{-1} H^{-1}, \tag{54}$$

equation (51) becomes

$$\frac{\partial \boldsymbol{r}}{\partial \boldsymbol{\omega}'} = -I + M^{-1} H^{-1}. \tag{55}$$

The own-country response is

$$\frac{\partial r_i}{\partial \omega_i} = -1 + \frac{[M^{-1}]_{ii}}{h_i}. \tag{56}$$

Since

$$M = H^{-1} + J \succeq H^{-1}, \tag{57}$$

the ordering of positive definite matrices implies

$$M^{-1} \preceq H. \tag{58}$$

Furthermore, $J_{ii} > 0$ ensures the strict diagonal inequality

$$0 < [M^{-1}]_{ii} < h_i. \tag{59}$$

Hence,

$$-1 < \frac{\partial r_i}{\partial \omega_i} < 0. \tag{60}$$

For $i \neq j$,

$$\frac{\partial r_j}{\partial \omega_i} = \frac{[M^{-1}]_{ji}}{h_i}. \tag{61}$$

Because M is positive definite with strictly negative off-diagonal elements, it is a nonsingular irreducible M-matrix. Its inverse is strictly positive entrywise.

Therefore,

$$\frac{\partial r_j}{\partial \omega_i} > 0 \qquad (j \neq i). \tag{62}$$

Combining the own- and cross-country responses,

$$\frac{\partial(r_i - r_j)}{\partial \omega_i} = \frac{\partial r_i}{\partial \omega_i} - \frac{\partial r_j}{\partial \omega_i} < 0. \tag{63}$$

This completes the proof.

B.2 Proof of Proposition 2

Proof. We first derive the equilibrium response of interest rates to destination attractiveness in the general N-country economy. We then characterize the response under symmetry.

Step 1. Portfolio allocation and capital supply. The portfolio share of investor n allocated to destination i is

$$b_{ni} = \frac{\exp[(r_i + \omega_i - \Phi_{ni})/\sigma_N]}{\sum_m \exp[(r_m + \omega_m - \Phi_{nm})/\sigma_N] + D_n^{ROW}}, \tag{64}$$

where $D_n^{ROW} \geq 0$ denotes the contribution of the outside option, if present, to the denominator. Define the attractiveness-adjusted return

$$q_i = r_i + \omega_i. \tag{65}$$

Capital supplied to destination i is

$$K_i^s = \sum_n A_n b_{ni}. \tag{66}$$

Differentiating with respect to q_j gives

$$\frac{\partial b_{ni}}{\partial q_j} = \frac{1}{\sigma_N} b_{ni} \left(\mathbf{1}\{i = j\} - b_{nj} \right). \tag{67}$$

Let J denote the Jacobian of capital supply:

$$J_{ij} \equiv \frac{\partial K_i^s}{\partial q_j} = \frac{1}{\sigma_N} \sum_n A_n b_{ni} \left(\mathbf{1}\{i = j\} - b_{nj} \right). \tag{68}$$

Equivalently,

$$J = \frac{1}{\sigma_N} \sum_n A_n \left[\operatorname{diag}(\boldsymbol{b}_n) - \boldsymbol{b}_n \boldsymbol{b}_n' \right], \tag{69}$$

where $\boldsymbol{b}_n$ contains the portfolio shares across the N domestic and foreign destinations, excluding the outside option.

The matrix J is symmetric and positive semidefinite. Under strictly positive portfolio shares,

$$J_{ii} > 0, \qquad J_{ij} < 0 \quad (i \neq j). \tag{70}$$

Step 2. Production and market clearing. The equilibrium interest rate satisfies

$$r_i = \alpha z_i \left(\frac{k_i}{\ell_i}\right)^{\alpha-1} - \delta. \tag{71}$$

Define

$$h_i \equiv -\frac{\partial r_i}{\partial k_i} = (1-\alpha)\frac{r_i + \delta}{k_i} > 0 \tag{72}$$

and

$$H = \operatorname{diag}(h_1, \ldots, h_N). \tag{73}$$

Differentiating the production condition yields

$$d\boldsymbol{r} = -H\, d\boldsymbol{k}. \tag{74}$$

Capital market clearing requires

$$\boldsymbol{k} = \boldsymbol{K}^s(\boldsymbol{r} + \boldsymbol{\omega}). \tag{75}$$

Holding investor asset stocks $\{A_n\}$ fixed, total differentiation gives

$$d\boldsymbol{k} = J(d\boldsymbol{r} + d\boldsymbol{\omega}). \tag{76}$$

Combining (73) and (75),

$$d\boldsymbol{r} = -HJ(d\boldsymbol{r} + d\boldsymbol{\omega}). \tag{77}$$

Rearranging,

$$(H^{-1} + J)d\boldsymbol{r} = -J\, d\boldsymbol{\omega}. \tag{78}$$

Hence,

$$d\boldsymbol{r} = -P\, d\boldsymbol{\omega}, \tag{79}$$

where

$$P = (H^{-1} + J)^{-1}J. \tag{80}$$

Step 3. Relative attractiveness and interest rates. Let

$$M = H^{-1} + J. \tag{81}$$

Because H^{-1} is positive definite and J is positive semidefinite, M is positive definite.

Furthermore, M has strictly negative off-diagonal elements and positive diagonal elements.

To establish its inverse-positivity, note that

$$\begin{aligned} M_{ii} - \sum_{j \neq i} |M_{ij}| = \frac{1}{h_i} \\ + \frac{1}{\sigma_N} \sum_n A_n b_{ni} \left(1 - \sum_m b_{nm}\right) > 0. \end{aligned} \tag{82}$$

Thus, M is a nonsingular irreducible M-matrix, and its inverse is strictly positive entrywise.

Using

$$P = M^{-1}J = I - M^{-1}H^{-1}, \tag{83}$$

we obtain, for $i \neq j$,

$$P_{ij} = -\frac{[M^{-1}]_{ij}}{h_j} < 0. \tag{84}$$

Moreover, since $J \succeq 0$,

$$M^{-1} \preceq H. \tag{85}$$

The inequality is strict on the diagonal because $J_{ii} > 0$. Consequently,

$$P_{ii} = 1 - \frac{[M^{-1}]_{ii}}{h_i} \in (0, 1). \tag{86}$$

Now consider two distinct destinations i and j. Define relative attractiveness as

$$\Delta\omega_{ij} = \omega_i - \omega_j. \tag{87}$$

Holding their average attractiveness and all other destinations' attractiveness fixed,

$$d\boldsymbol{\omega} = \frac{1}{2}(e_i - e_j) d\Delta\omega_{ij}, \tag{88}$$

where e_i denotes the i-th unit vector.

It follows from (78) that

$$\begin{aligned} d(r_i - r_j) &= -(e_i - e_j)' P \, d\boldsymbol{\omega} \\ &= -\frac{1}{2}(e_i - e_j)' P (e_i - e_j) d\Delta\omega_{ij}. \end{aligned} \tag{89}$$

Therefore,

$$\frac{\partial(r_i - r_j)}{\partial(\omega_i - \omega_j)} = -\frac{1}{2}\left(P_{ii} + P_{jj} - P_{ij} - P_{ji}\right) < 0. \tag{90}$$

The inequality follows from $P_{ii}, P_{jj} > 0$ and $P_{ij}, P_{ji} < 0$.

Step 4. Symmetric multi-country equilibrium. Suppose the reference equilibrium is symmetric across all N destinations.

In particular, the curvature of the production condition is identical across destinations:

$$H = hI, \qquad h > 0. \tag{91}$$

Suppose further that the capital-supply Jacobian is invariant under permutations of destination countries.

By permutation symmetry, J admits the representation

$$J = \lambda_\perp \left(I - \frac{\mathbf{1}\mathbf{1}'}{N}\right) + \lambda_\parallel \frac{\mathbf{1}\mathbf{1}'}{N}, \tag{92}$$

where $\lambda_\perp > 0$ and $\lambda_\parallel \geq 0$.

The first component captures the response to relative changes in destination attractiveness, while the second captures the response to a common change across all destinations.

Substituting (91) into (79) yields

$$P = \kappa \left(I - \frac{\mathbf{1}\mathbf{1}'}{N}\right) + \kappa_\parallel \frac{\mathbf{1}\mathbf{1}'}{N}, \tag{93}$$

where

$$\kappa = \frac{h\lambda_\perp}{1 + h\lambda_\perp} \in (0, 1) \tag{94}$$

and

$$\kappa_\parallel = \frac{h\lambda_\parallel}{1 + h\lambda_\parallel}. \tag{95}$$

Since

$$(e_i - e_j)'\mathbf{1} = 0, \tag{96}$$

we have

$$(e_i - e_j)'P = \kappa(e_i - e_j)'. \tag{97}$$

Consequently, for any local change in the attractiveness vector,

$$d(r_i - r_j) = -\kappa\, d(\omega_i - \omega_j), \qquad 0 < \kappa < 1. \tag{98}$$

This completes the proof.

□

B.3 Proof of Proposition 3

Proof. Consider the symmetric N-country reference equilibrium characterized in Proposition 2. Let destination attractiveness vary around its symmetric reference value according to

$$\omega_i(\epsilon) = \bar{\omega} + \epsilon u_i, \tag{99}$$

where u_i is a random perturbation with finite second moments.

Define the first-order equilibrium interest-rate response as

$$\dot{r}_i = \left.\frac{dr_i(\epsilon)}{d\epsilon}\right|_{\epsilon=0}. \tag{100}$$

By the symmetric-equilibrium result of Proposition 2,

$$\dot{r}_i - \dot{r}_j = -\kappa(u_i - u_j), \qquad 0 < \kappa < 1. \tag{101}$$

Taking the conditional variance on both sides yields

$$\operatorname{Var}(\dot{r}_i - \dot{r}_j \mid d_{ij}) = \kappa^2 \operatorname{Var}(u_i - u_j \mid d_{ij}). \tag{102}$$

Since κ is constant across country pairs in the symmetric equilibrium, differentiation with respect to logarithmic distance gives

$$\begin{aligned} &\frac{\partial \operatorname{Var}(\dot{r}_i - \dot{r}_j \mid d_{ij})}{\partial \ln d_{ij}} \\ &\quad = \kappa^2 \frac{\partial \operatorname{Var}(u_i - u_j \mid d_{ij})}{\partial \ln d_{ij}}. \end{aligned} \tag{103}$$

Therefore, a positive geographic gradient in the variance of attractiveness differences implies a positive geographic gradient in the variance of equilibrium interest-rate differentials.

Furthermore, taking absolute values in equation (**??**) and then conditional expectations yields

$$\mathbb{E}[|\dot{r}_i - \dot{r}_j| \mid d_{ij}] = \kappa \mathbb{E}[|u_i - u_j| \mid d_{ij}]. \tag{104}$$

Consequently,

$$\begin{aligned} &\frac{\partial \mathbb{E}[|\dot{r}_i - \dot{r}_j| \mid d_{ij}]}{\partial \ln d_{ij}} \\ &\quad = \kappa \frac{\partial \mathbb{E}[|u_i - u_j| \mid d_{ij}]}{\partial \ln d_{ij}}. \end{aligned} \tag{105}$$

Thus, an increasing expected absolute difference in destination attractiveness also implies an increasing expected absolute equilibrium interest-rate differential.

Both results hold without imposing Gaussianity or a parametric spatial covariance function.

□

C Additional Figures

C.1 Model Fit

Figure 2 compares the model-implied interest rates with their empirical counterparts. The model reproduces the overall cross-country variation in interest rates.

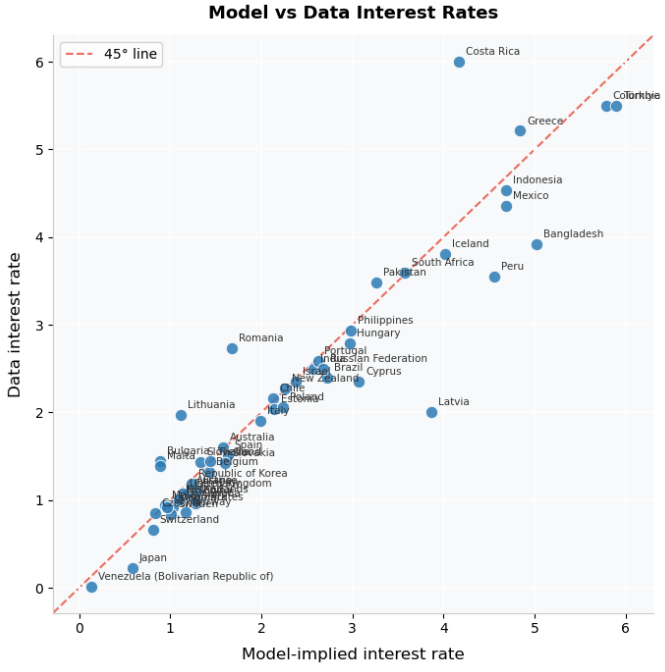
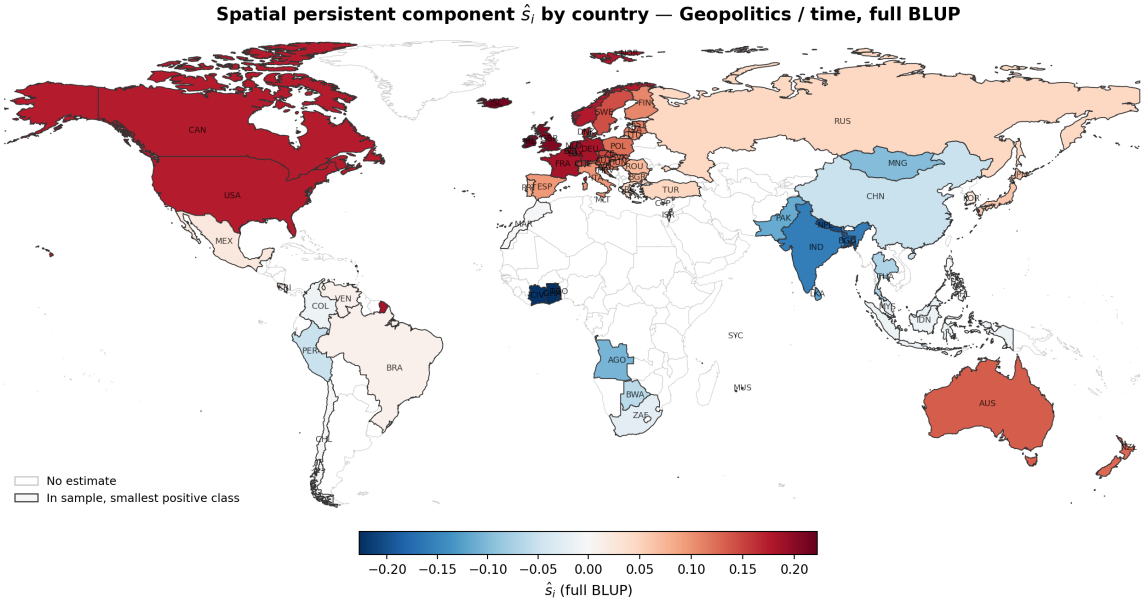


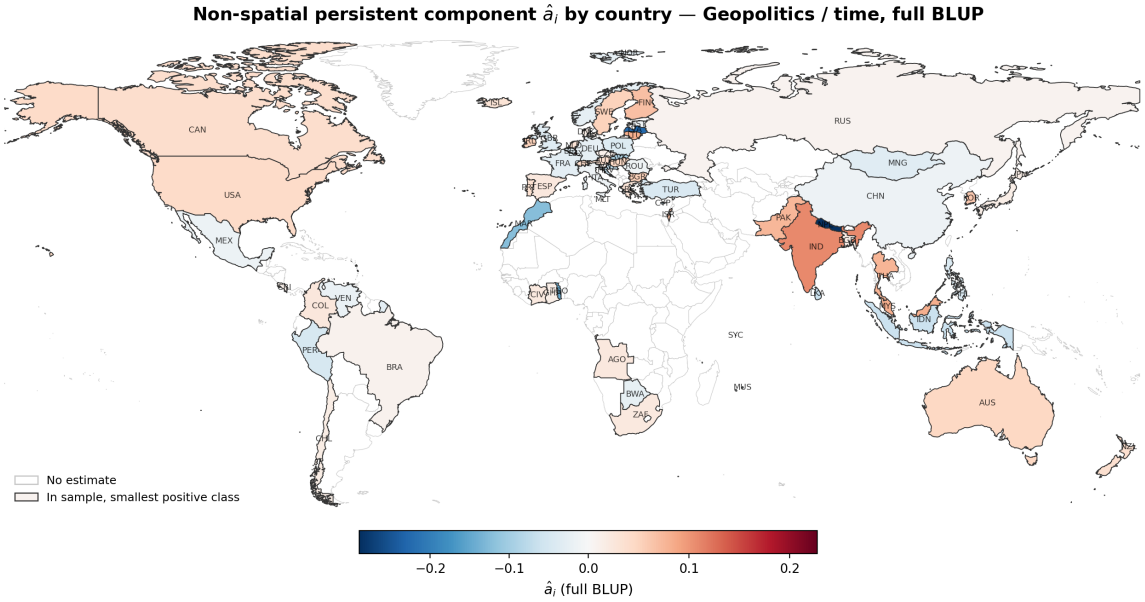
Figure 2: Model-Implied and Observed Interest Rates

C.2 Spatial Decomposition of Destination Attractiveness

Figure 3 presents the estimated persistent components of destination attractiveness. Panel (a) shows the spatial component, $\hat{s}_i$, while Panel (b) shows the non-spatial country-specific component, $\hat{a}_i$. Both components are recovered using the full-sample BLUP estimates described in Section 6.4.



(a) Spatial persistent component, $\hat{s}_i$



(b) Non-spatial persistent component, $\hat{a}_i$

Figure 3: Decomposition of Destination Attractiveness

C.3 Counterfactual Reallocation with Fixed Asset Stocks

Figure 4 shows how capital is reallocated across destinations when each geographic channel is removed while country-level asset stocks are held fixed at their baseline values. Conditional on these asset stocks, portfolio shares, destination capital stocks, and equilibrium interest rates are jointly re-solved.

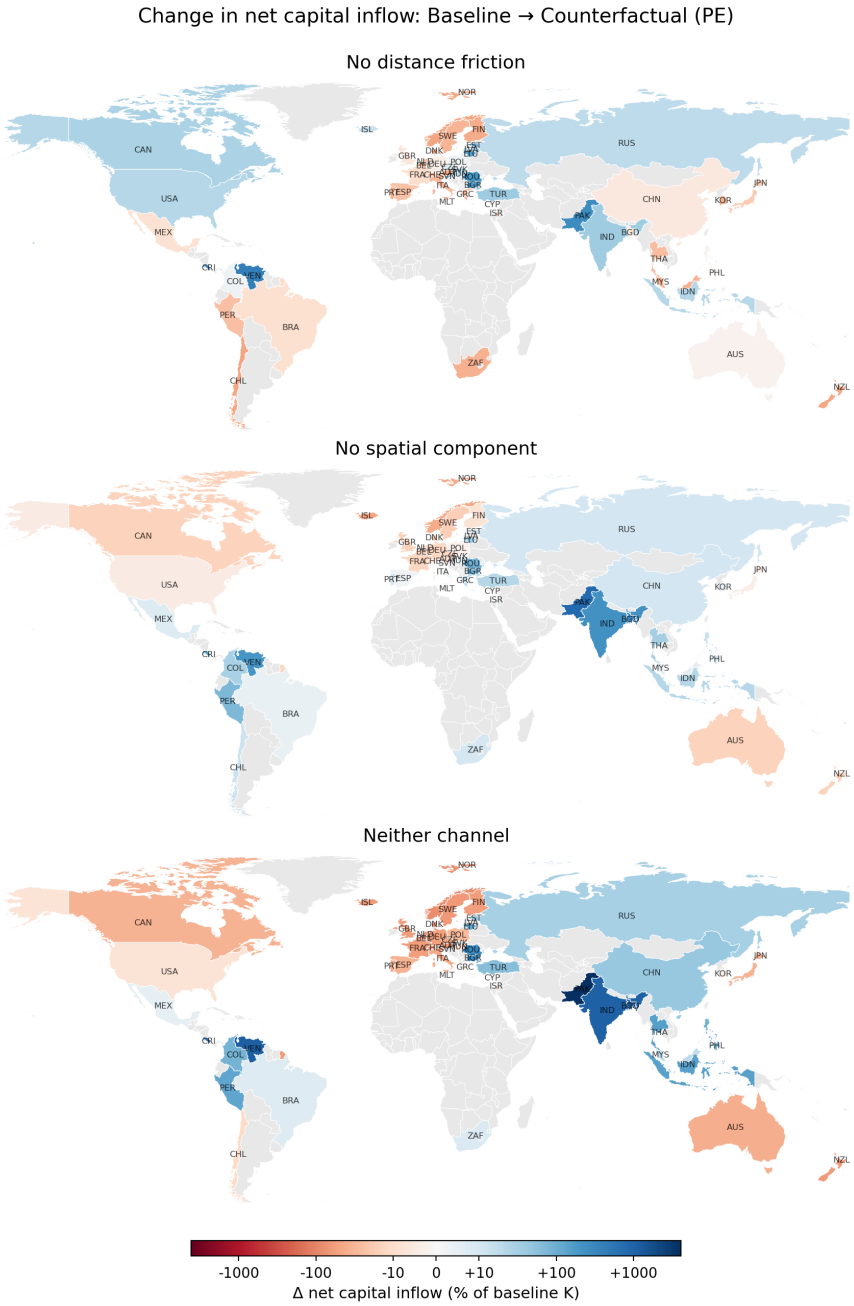


Figure 4: Capital Reallocation with Fixed Asset Stocks

Figure 5 reports the corresponding changes in interest rates. Each point compares the baseline interest rate with the counterfactual interest rate for the same country. Points above the 45-degree line indicate an increase in the equilibrium interest rate relative to the baseline, while points below the line indicate a decrease.

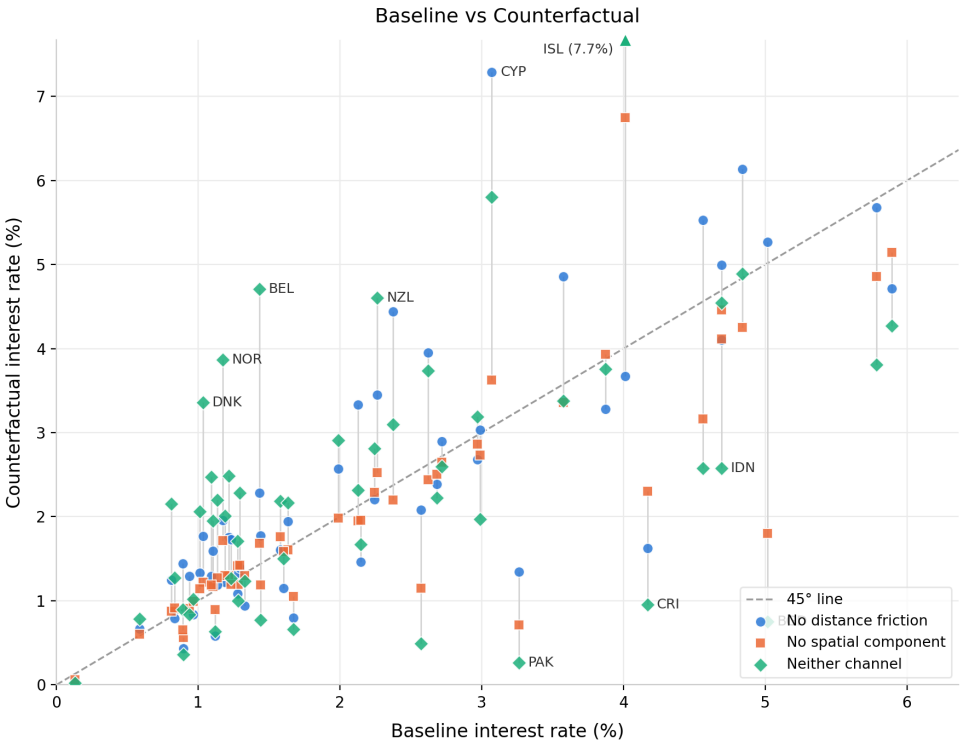


Figure 5: Interest-Rate Responses with Fixed Asset Stocks

D Robustness Checks

We examine whether the estimated effects of geographic distance and geopolitical alignment on portfolio allocation are robust to alternative specifications. Recall that the bilateral wedge ϕ_{ni} enters portfolio demand with a negative sign. Thus, a negative coefficient in the portfolio allocation equation corresponds to a larger portfolio wedge.

First, we augment the baseline specification with additional bilateral covariates, including exchange-rate volatility, common language, and trade intensity. Table 8 reports the baseline estimate in the full sample, the baseline estimate in the common sample, and specifications that sequentially add these controls. The coefficients on log geographic distance and the cross-block indicator remain negative and statistically significant across specifications.

Table 8: Robustness of Bilateral Portfolio Friction Estimates

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(dist_i,j)$	-0.797***	-0.790***	-0.716***	-0.726***	-0.706***	-0.650***
	(0.054)	(0.055)	(0.052)	(0.058)	(0.063)	(0.062)
Foreign	-4.159***	-4.171***	-3.730***	-4.424***	-4.478***	-3.991***
	(0.130)	(0.131)	(0.117)	(0.144)	(0.156)	(0.148)
Cross Block	-0.796***	-0.803***	-0.705***	-0.773***	-0.819***	-0.722***
	(0.155)	(0.157)	(0.172)	(0.156)	(0.157)	(0.171)
FX volatility (%)			-0.288***			-0.279***
			(0.030)			(0.030)
Common language				0.661***		
				(0.169)		
Trade intensity (%)					0.141***	0.108***
					(0.039)	(0.036)
Sample	Full	Common	Common	Common	Common	Common
Observations	100377	95086	95086	95086	95086.000	95086.000

Standard errors in parentheses

Origin×year and destination×year fixed effects. Standard errors clustered by country pair.

FX volatility: s.d. of monthly bilateral exchange-rate returns within the year (%).

Trade intensity: (X_ij+X_ji)/(GDP_i+GDP_j) (%), IMF IMTS and World Bank.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We also allow home bias to vary across investor countries. Specifically, we parameterize the bilateral wedge as

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta_n^{foreign} \mathbb{1}[n \neq i] + \beta^{cross\text{-}block} CrossBlock_{ni}.$$

Allowing for investor-specific home bias leaves the main results unchanged. The coefficients on log

geographic distance and the cross-block indicator in the portfolio allocation equation remain negative and statistically significant.

Table 9: Robustness to Investor-Specific Home Bias

	(1) share
Log distance	-0.921***
	(0.045)
Cross block	-0.579***
	(0.216)
Observations	100377.000

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

E ω Regression

Table 10 examines how the recovered destination attractiveness, $\omega_{i,t}$, is related to observable country characteristics. By construction, $\omega_{i,t}$ captures components of destination attractiveness other than interest rates.

Across specifications, real GDP is positively associated with $\omega_{i,t}$, suggesting that larger markets attract greater portfolio allocation conditional on interest rates. In contrast, CDS spreads are negatively associated with $\omega_{i,t}$, indicating that investors allocate smaller portfolio shares to countries with greater sovereign credit risk, holding interest rates fixed. These relationships remain present when country fixed effects are included, so they are not driven solely by persistent cross-country differences.

Political stability is positively associated with destination attractiveness in some specifications, although the estimates are less precise.

Table 10: Destination Attractiveness and Country Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
ln(Population)	0.055***		0.059***	-0.439**		-0.406*
	(0.012)		(0.011)	(0.204)		(0.206)
Sovereign Credit Score		0.030***	0.034***		0.011**	0.008*
		(0.008)	(0.006)		(0.005)	(0.004)
Country FE	No	No	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	531	531	531	528	528	528
Countries	66	66	66	63	63	63
Adj. R^2	0.281	0.132	0.473	0.960	0.958	0.960

Standard errors clustered by country in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Latent Attractiveness with Geographic and Geopolitical Dependence

Table 4b provides evidence that countries' estimated latent attractiveness is associated with both geographic and geopolitical proximity, with the latter measured by distance in UN General Assembly voting ideal points. To capture these two sources of dependence, we specify an additive Gaussian covariance kernel:

$$\mathrm{Cov}(s_i, s_j) = \sigma_G^2 \exp\left[-\left(\frac{\tilde{d}_{G,ij}}{\rho_G}\right)^2\right] + \sigma_P^2 \exp\left[-\left(\frac{\tilde{d}_{P,ij}}{\rho_P}\right)^2\right]. \tag{106}$$

Here, the subscripts G and P denote the geographic and geopolitical components, respectively. Geographic distance is measured by chordal distance, while geopolitical distance is the absolute difference between countries' sample-period average ideal points. Each distance is divided by its standard deviation across distinct unordered country pairs, without subtracting its mean. The parameters σ_G^2 and σ_P^2 govern the component variances, while ρ_G and ρ_P govern their correlation ranges.

This specification corresponds to the decomposition $s_i = s_i^G + s_i^P$, with $\mathrm{Cov}(s_i^G, s_j^P) = 0$ for all i, j.

This restriction concerns the latent components and does not require geographic and geopolitical distances to be uncorrelated. We also retain the independent country-specific component a_i, with variance σ_a^2.

Using the geopolitical specification of ω_{it}, with 1,028 country-year observations from 67 countries and controlling for log population and year fixed effects, we estimate $\hat{\sigma}_G^2 = 0.00552$, $\hat{\sigma}_P^2 = 0.01501$, and $\hat{\sigma}_a^2 = 0.00791$. These components account for approximately 19.4%, 52.8%, and 27.8% of the model-implied persistent variance, respectively. The estimated decomposition therefore attributes a substantial share of persistent heterogeneity to geopolitical dependence, while retaining geographic dependence and an independent country-specific component.

Converted to the original distance units, the estimated geographic and geopolitical ranges are 1,677 km and 1.366 ideal-point units, respectively. At these distances, the correlation within the corresponding component falls to e^{-1}, approximately 0.37. The geographic range is shorter than the 2,929 km estimated under the geography-only covariance specification, indicating a reallocation of covariance when geopolitical dependence is modeled separately.

A parametric bootstrap test with 499 replications rejects the null hypothesis $H_0 : \sigma_P^2 = 0$ at the 5% level ($p = 0.012$). This result provides evidence of geopolitical dependence beyond that captured by the geographic covariance component.